Further derivatives of functions — Problem Set

Warm-up

1. Find $\frac{d}{dx}(\sin^{-1} x)$.
2. Differentiate $y = \cos^{-1} x$.
3. Find the derivative of $f(x) = \tan^{-1} x$.
4. Find $\frac{dy}{dx}$ if $y = \sin^{-1}(5x)$.
5. Differentiate $y = \tan^{-1}(2x)$.
6. Find the derivative of $y = \cos^{-1}(kx)$ where k is a constant.
7. Differentiate $f(x) = \sin^{-1}(\frac{x}{3})$.
8. Find $f'(x)$ if $f(x) = \tan^{-1}(\frac{x}{2})$.
9. Differentiate $y = 3\cos^{-1} x$.
10. Find $\frac{d}{dx}[\sin^{-1}(x^2)]$.

Skill-building

1. Differentiate $y = x\tan^{-1} x$ using the product rule.
2. Find the derivative of $f(x) = \sqrt{\sin^{-1} x}$.
3. Use the chain rule to find $\frac{dy}{dx}$ for $y = \cos^{-1}(e^x)$.
4. Differentiate $y = \frac{\tan^{-1} x}{x}$.
5. Find the derivative of $y = \sin^{-1}(2x + 1)$.
6. Given $y = \cos^{-1}(\frac{1}{x})$, show that $\frac{dy}{dx} = \frac{1}{x\sqrt{x^2-1}}$ for $x > 1$.
7. Differentiate $f(x) = (\tan^{-1} x)^2$.
8. Find $\frac{dy}{dx}$ if $y = \sin^{-1}(\cos x)$.
9. Given $f(x) = x^3 + x$, find the derivative of the inverse function $f^{-1}(x)$ at $x = 2$.
10. Differentiate $y = \tan^{-1}(\sqrt{x})$.

Easier Exam Questions

1. (Cherrybrook Tech 2020 Q4) What is the derivative of $\sin^{-1} 3x$?

   (A) $-\frac{1}{\sqrt{1-9x^2}}$

   (B) $\frac{3}{\sqrt{1-9x^2}}$

   (C) $-\frac{1}{\sqrt{1-3x^2}}$

   (D) $-\frac{3}{\sqrt{1-9x^2}}$

2. (Girraween 2021 Q6) What is the derivative of $y = \cos^{-1}\left(\frac{x}{4}\right)$

   (A) $-\frac{1}{\sqrt{16-x^2}}$

   (B) $-\frac{2}{\sqrt{16-x^2}}$

   (C) $-\frac{4}{\sqrt{16-x^2}}$

   (D) $-\frac{6}{\sqrt{16-x^2}}$

3. (Cheltenham Girls 2022 Q1) What is the derivative of $\tan^{-1}(x^4)$ with respect to x?

   (A) $\frac{1}{1+x^8}$

   (B) $\frac{4x^3}{1+x^8}$

   (C) $\frac{4x^3}{1+x^4}$

   (D) $\frac{x^4}{1+x^8}$

4. (The Kings School 2021 Q4) Which of the following is the derivative of $y = x^2 \sin^{-1}(2x)$?

   (A) $2x\sin^{-1}(2x) + \frac{2x^2}{\sqrt{1-4x^2}}$

   (B) $2x\sin^{-1}(2x) + \frac{x^2}{\sqrt{1-4x^2}}$

   (C) $2x\sin^{-1}(2x) + \frac{2x^2}{\sqrt{1-2x^2}}$

   (D) $2x\sin^{-1}(2x) + \frac{x^2}{\sqrt{1-2x^2}}$

5. (Cheltenham Girls 2020 Q6) If $y = \sin^{-1}\frac{a}{x}$ then $\frac{dy}{dx}$ equals:

   (A) $\frac{-a}{x^2\sqrt{x^2-a^2}}$

   (B) $\frac{x}{\sqrt{x^2-a^2}}$

   (C) $\frac{-x}{\sqrt{x^2-a^2}}$

   (D) $\frac{-a}{x\sqrt{x^2-a^2}}$

Harder Exam Questions

1. (Sydney Girls 2024 Q11a) Differentiate $(\sin^{-1} x)^3$. **2**

2. (Cheltenham Girls 2020 Q5) Given that $f(x) = e^x - 1$ and $y = f^{-1}(x)$, find an expression for $\frac{dy}{dx}$.

(A) $\frac{1}{e^x-1}$

(B) $\frac{1}{x+1}$

(C) $\ln x$

(D) $\ln(x+1)$

3. (Manly 2022 Q4) The derivative of $y = \cos^{-1}[2f(x)]$ is :

   (A) $\frac{dy}{dx} = -\frac{1}{\sqrt{1-[f(x)]^2}}$

   (B) $\frac{dy}{dx} = -\frac{2}{\sqrt{1-4[f(x)]^2}}$

   (C) $\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1-4[f(x)]^2}}$

   (D) $\frac{dy}{dx} = -\frac{2f'(x)}{\sqrt{1-4[f(x)]^2}}$

4. (Fort Street 2020 Q11b) Show that the derivative of $y = \frac{1}{2}\cos^{-1}(3x+1)$ is given by **2**

$$y' = -\frac{1}{2x}\sqrt{\frac{-3x}{3x+2}}$$

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5. (Trinity Grammar 2022 Q12b)

   (i) Differentiate $\tan^{-1}\left(\frac{1}{x}\right)$, $x \neq 0$, and hence show that **2**

$$\frac{d}{dx}\left(\tan^{-1} x + \tan^{-1}\left(\frac{1}{x}\right)\right) = 0.$$

   (ii) Hence explain why $\tan^{-1} x + \tan^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2}$ for $x > 0$. **2**